

1. What is Power BI?

Power BI is a business intelligence and data visualization tool developed by Microsoft. It helps to connect to multiple data sources, transform the data, and create interactive reports & dashboards.

2. Why should we use Power BI?

We use Power BI because it connects to many data sources, provides drag-and-drop visualizations, supports real-time data updates, offers DAX for calculations, enables collaboration via Power BI Service, and is cost-effective.

3. Difference between Power BI and Tableau?

Power BI (Microsoft) is easier for beginners, cheaper, integrates with MS products, and best for small/medium businesses. Tableau (Salesforce) is more flexible, expensive, requires training, and is better for complex/large-scale organizations.

4. What is Power BI Desktop?

Power BI Desktop is a free Windows application to connect to data, clean/transform using Power Query, build models, and create dashboards. Reports can be published to Power BI Service.

5. What is DAX?

DAX (Data Analysis Expressions) is a formula language used in Power BI, Excel, and Analysis Services to create calculated columns, measures, and tables. Examples: SUM(), CALCULATE(), FILTER(), RELATED().

6. What are Filters in Power BI?

Filters refine data in reports. Types: Visual Level (single chart), Page Level (all visuals in a page), Report Level (all pages), and Drill-through Filters (context-based filtering).

7. Explain how relationships are defined in Power BI Desktop.

Relationships connect tables using keys. You define cardinality (One-to-One, One-to-Many, Many-to-Many), cross-filter direction (Single/Both), and active/inactive relationships in Model View.

8. What is the CALCULATE function in DAX?

CALCULATE() modifies the filter context of a calculation. Syntax: CALCULATE(, , ...) Example: TotalSales = CALCULATE(SUM(Sales[Amount]), Sales[Region] = 'West') This calculates sales only for the West region. It is used for advanced metrics like YTD, conditional totals, and rolling averages.